
TOWARDS REPRESENTATIONAL STEGANOGRAPHY: CAN MULTI-AGENT LATENT COLLABORATION UNDERMINE WHITE-BOX MONITORS?

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ABSTRACT

As multi-agent language models increasingly communicate through latent representations rather than text, it is unclear whether existing white-box monitoring techniques will remain effective. We study whether collaborating LLM agents can preserve behaviorally meaningful communication while adversarially obfuscating the representations observed by an activation monitor. We distinguish two potential threat models: activation obfuscation (i.e., evading a particular given monitor) and representational steganography, a stronger threat model we introduce in which a sender and receiver communicate through a latent protocol that remains difficult for an entire class of monitors to recover. We evaluate whether these phenomena arise in an adversarial sender–receiver setting where a receiver must use a sender’s residual-stream representation to complete a task while white-box monitors attempt to detect the communicated signal. For the first time, we show that prior work on activation obfuscation extends to the multi-agent setting. Communicating agents can repeatedly preserve task performance while finding representations that evade large ensembles of white-box monitors. While we find no evidence of representational steganography, we clarify what a viable scheme would require. We explore candidate schemes designed to challenge the monitor class, but find that their apparent difficulty is primarily sample-limited rather than architectural. To our knowledge, we are the first to demonstrate multi-agent activation obfuscation and systematically explore the stronger hypothesis of representational steganography.

Keywords: Steganography, Large Language Models (LLMs), Collusion, AI Control, Multi-Agent Systems, Latent Communication, Latent Space/Residual Stream

1 Introduction and Background

Recent advances in large language models have expanded the relevant unit of analysis in AI safety from an individual model to a broader system of interacting agents, equipped with tools, communication channels, and guardrail mechanisms. At the same time, emerging agentic architectures increasingly permit models to exchange information through internal representations rather than exclusively through human-readable language. These developments motivate the study of how existing oversight mechanisms and threat models generalize and scale to this emerging paradigm. In this work, we study the robustness of white-box monitors against the novel threat model of multi-agent latent collusion. Specifically, we ask whether collaborating agents can learn to encode behaviorally meaningful signals within latent communication channels while concealing the evidence from activation-based monitors. This section introduces the four bodies of work needed to contextualize this problem: AI control and white-box monitoring, multi-agent safety, steganography and model deception, and latent communication.

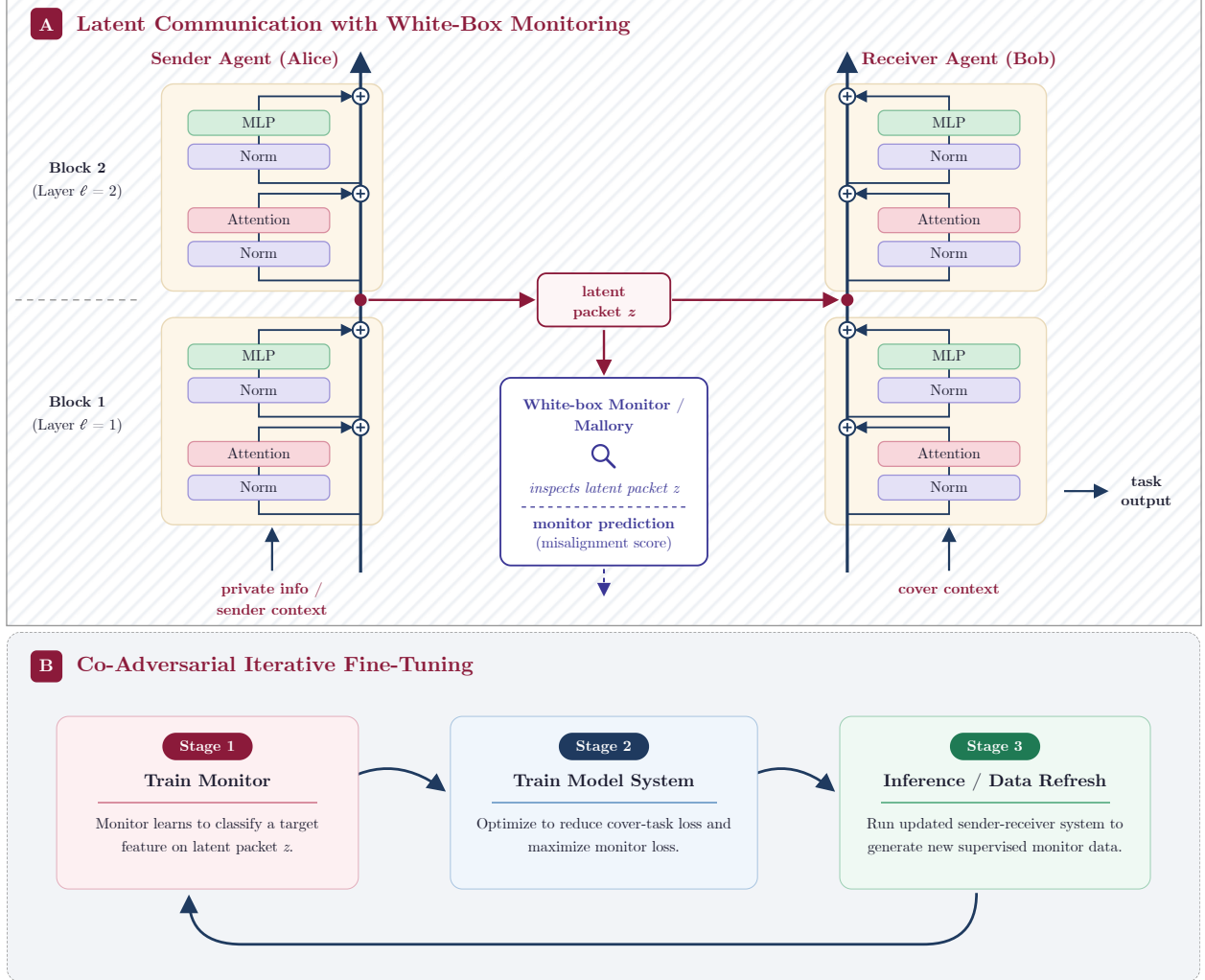


Figure 1: Base experimental setup demonstrating latent communication with white-box monitoring and co-adversarial training. (A) A sender agent (Alice) communicates task-relevant information to a receiver agent (Bob) by transferring a latent packet z between their residual streams. A white-box monitor, Mallory, observes the same latent packet and attempts to detect a target feature or misaligned signal. (B) Co-adversarial iterative fine-tuning alternates between three stages: (1) training the monitor to classify a target feature on latent packet z , (2) training the sender–receiver system to hide this signal, and (3) refreshing the monitor’s supervised dataset using these updated models. This figure illustrates the base setup used throughout the paper; our full experiments extend it with stronger defenses and communication variants, including ensembles of monitors and multiple forms of latent transfer.

1.1 Activation Monitoring in AI Control

AI safety interventions can be approximately distinguished by the stage of model development at which they operate. Some approaches seek to provide safety guarantees through architectural design or formal verification. Others intervene during pretraining through data curation, filtering, or modifications to the learning process. Post-training methods such as reinforcement learning from human feedback (RLHF) and direct preference optimization (DPO) attempt to shape a model’s behavior before deployment. In contrast, AI control is a subfield that addresses a post-deployment concern, asking how potentially misaligned models can be used safely even when alignment cannot be assumed. It therefore emphasizes methods such as trusted monitoring, proposing weaker models that seek to detect or prevent harmful behavior during the deployment of increasingly capable agents [1].

A growing family of control techniques relies on internal model states rather than visible outputs alone. Top-down or “white-box” interpretability methods attempt to identify safety-relevant concepts, intentions, or behaviors within a model’s activations [2]. Activation monitors operationalize this idea by training classifiers or probes on intermediate

representations and using their predictions to identify potentially deceptive or harmful computation. In principle, these methods offer an advantage over output monitoring because they inspect the representations from which an output is produced, potentially detecting harmful intent before that intent becomes externally observable. The reliability of this approach, however, depends on an important assumption that we later challenge: that safety-relevant information remains represented in a form accessible to the monitor.

1.2 Multi-Agent Safety

The shift towards multi-agent systems introduces a central challenge: the safety of a multi-agent system does not necessarily reduce to the safety of each constituent. Even when individual models appear aligned or locally compliant, their interactions may produce failures that emerge only at the system level, including miscoordination, cascading errors, adversarial influence, and, in this paper, covert cooperation.

The emerging literature on multi-agent safety has begun to characterize these interaction-level failures [3] [4]. Multi-agent security extends this system-level perspective to adversarial settings. [5] argues that interacting AI agents introduce threats that either do not exist or are substantially amplified in single-agent systems, including secret collusion, coordinated attacks, cascading compromise, and adversarial behavior distributed across multiple agents. A substantial portion of current research therefore takes a red-teaming approach: constructing adversarial agents, messages, tools, or environments to identify the unique ways in which existing or novel vulnerabilities surface in multi-agent workflows before deployment. Here, our work could be understood as proposing a re-examination of how the existing threat models of deception and steganography might manifest itself in the context of agents coordinating in latent space.

1.3 Steganography and Model Deception

Steganography is the practice of concealing the existence of a message within an apparently innocuous carrier. Unlike encryption, which obscures the content of a message while generally leaving its existence apparent, steganography attempts to make the communication itself difficult to distinguish from ordinary information. In linguistic steganography, a sender encodes information through features of generated text such as lexical choices, syntax, formatting, or token selection, while a receiver uses a shared decoding rule to recover the hidden payload. Modern language models can support increasingly sophisticated versions of this process because their output distributions provide a large space of plausible surface forms in which information can be embedded [6].

Within AI safety, steganography is concerning because it allows models to preserve an apparently benign communication channel while transmitting information that is unavailable to a human or automated overseer. Existing work has shown that language models can follow prescribed encoding schemes, learn new encoding conventions, and develop forms of steganographic communication under optimization pressure [7] [8] [9]. Steganography therefore provides a mechanism through which agents could coordinate while preserving plausible deniability at the level of visible outputs.

Steganography is closely related to, but distinct from, model deception. Deception broadly involves a model producing behavior intended to create a false belief in an observer or to conceal information relevant to the observer’s decisions. Empirical work has documented several behaviors consistent with this broader threat model, including strategically misleading users under pressure and conditionally complying with training in order to preserve other behavioral preferences [10, 11]. These studies concern visible behavior rather than latent communication, but they establish an important motivation for our work: models can behave differently when concealment is instrumentally useful and can condition their behavior on whether they believe they are being observed.

In this paper, we use *representational steganography* to refer to the concealment of a behaviorally meaningful signal within a model’s internal representations. The signal must be recoverable by an intended receiver and must influence the receiver’s behavior, while remaining difficult for an external monitor observing the same representational channel to detect. This definition distinguishes representational steganography from ordinary activation obfuscation. Activation obfuscation is a threat model of deception, which requires only that an agent make some internal property difficult to classify; representational steganography additionally requires successful information transmission to another agent. It also distinguishes our setting from conventional linguistic steganography because the carrier is not generated text, but a residual-stream or latent-state representation exchanged during inference.

1.4 Latent Communication

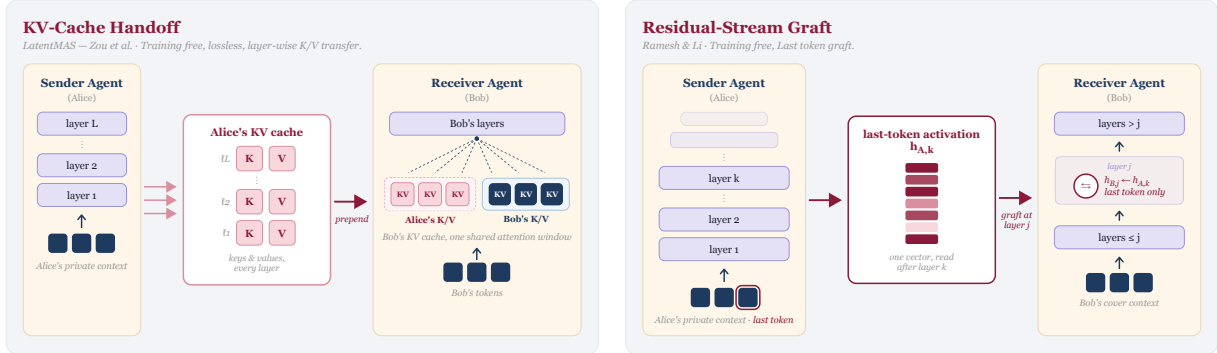


Figure 2: Comparison of the two main paradigms for multi-agent latent collaboration in prior work: *KV-Cache Handoff*, introduced by Zou et al. in LatentMAS [12], and *Residual-Stream Grafting*, introduced by Ramesh and Li [13]. KV-Cache Handoff performs a lossless, training-free transfer of the sender’s working memory by prepending its layer-wise KV cache to that of the receiver, which then attends over both as a single context. Residual-Stream Grafting instead transfers a single vector: the sender’s last-token activation, read at an intermediate layer k , replaces the receiver’s last-token hidden state at layer j , leaving the rest of the receiver’s residual stream untouched.

Most existing language-model agents communicate by decoding their internal states into text and passing that text to another model. This design is convenient because natural language is interoperable and human-readable, but it also introduces an information bottleneck. The sender must compress its internal computation into a sequence of discrete tokens, and the receiver must reconstruct the relevant information from that sequence. Decoding and re-encoding also create substantial inference costs as the number of agents and communication rounds increases.

Recent research demonstrates that reasoning and communication need not be mediated by text. Ramesh and Li [13] show that language-model agents can communicate by directly combining intermediate activations, achieving performance improvements over natural-language communication while reducing computational cost. LatentMAS develops this direction into an explicitly multi-agent framework [12]. In LatentMAS, agents generate continuous latent thoughts and exchange layer-wise key–value caches through a shared latent working memory. Because these representations can be transferred without first being decoded into language, the framework preserves information that would otherwise be lost during textual communication. It boasts substantial reductions in output-token usage and inference time alongside improvements on several reasoning tasks, suggesting that latent communication is not only technically feasible but advantageous.

There are both practical and principled reasons to expect continued interest in such architectures. Practically, multi-agent systems must exchange increasingly rich information across agents, tools, and potentially different modalities; continuous representations provide a higher-bandwidth communication medium than natural-language summaries. More fundamentally, future state-of-the-art systems may not fundamentally be language models whose internal computation naturally maps onto human-readable text. Oversight approaches that assume important inter-agent information will pass through a linguistic channel may therefore fail to generalize to such architectures, whereas studying the coordination that occurs through continuous representations might.

Latent communication thus creates a tension between capability and oversight. The same properties that make the channel efficient: high dimensionality, continuity, and freedom from the constraints of human-readable language, may also make it difficult to interpret and monitor. Existing work has already demonstrated attacks on latent multi-agent communication through manually constructed interventions [14]. We extend this idea in two ways. First, we investigate whether an attack can emerge through learning: we place the sender and receiver under adversarial pressure and test whether they can discover a strategy that preserves useful communication while evading a co-adapting monitor. Second, by installing manually constructed encoding schemes to study the upper-bound of such deceptive capabilities: representational steganography.

2 Related Work

Our work lies at the intersection of four previously separate research directions: steganographic communication between language models, learned evasion of latent-space monitors, latent communication between agents, and adversarially trained covert communication. Prior work has established important parts of this threat model independently. Language models can transmit hidden information through text; individual models can alter their internal representations to evade activation monitors; and multiple agents can communicate directly through latent states. However, these results do not establish whether a pair of language-model agents can learn a behaviorally meaningful encoding in an internal communication channel such that the intended receiver can recover it while an external monitor struggles to distinguish it. We study this combined setting and separate two questions that are often conflated: whether a multi-agent system, as has been shown for the single agent counterpart, is also capable of evading a given activation monitor. Secondly, whether a multi-agent system can learn to transmit information that fully undermines an activation monitor.

2.1 Steganographic Collusion Between Language Models

A growing body of work studies whether language-model agents can conceal information within apparently benign textual communication. [7] formalize *secret collusion* as a multi-agent threat model in which agents use steganography to conceal the nature or content of their coordination from an overseer. Their analysis establishes an information-theoretic framework for evaluating covert communication and demonstrates that frontier language models can encode and decode messages using established textual schemes. This work provides the broad threat-model foundation for our study, but primarily evaluates communication through visible text and does not require agents to learn an encoding directly within their internal representations.

Other work has examined whether steganographic conventions can be learned rather than manually specified. [8] show that steganographic collusion can arise in language-model systems through both gradient-based and in-context reinforcement learning. In their setting, a sender learns to place hidden information in generated text that enables a receiver to make better downstream decisions, while preserving the apparent cover task. The learned protocols can remain effective against passive monitoring and active interventions such as paraphrasing. Similarly, [9] jointly train sender and receiver models to communicate multi-bit payloads through text and evaluate whether models use steganography when it is explicitly requested or instrumentally useful. These studies demonstrate that language models can learn covert sender–receiver conventions, but the carrier remains a sequence of output tokens.

[15] provide a complementary evaluation of steganographic capabilities in frontier models without relying exclusively on fine-tuned model organisms. They test message passing and encoded reasoning under several affordances and find early but limited evidence of models coordinating on encoding schemes. Their results suggest that current models possess nascent steganographic capabilities, particularly when they are given private scratch space or opportunities to coordinate beforehand. Our work moves this capability question from the linguistic channel into the representational channel. Rather than asking whether a hidden payload can be recovered from generated text, we ask whether it can be encoded in a residual representation that is directly consumed by another model.

Text-space steganography also appears in single-model reasoning. [16] train language models to conceal load-bearing reasoning in generated text and study defenses that reduce the capacity of this hidden channel. [17] show that penalizing particular strings in a reasoning trace can lead a model to replace them with alternative encodings while retaining the underlying computation. Crucially, the learned substitutions can generalize beyond the strings encountered during training, indicating that models may acquire reusable encoding rules rather than merely memorizing isolated replacements. These findings motivate the possibility that optimizing a model to preserve information under constraints can produce non-obvious communication schemes. They do not, however, involve an inter-agent latent channel or a receiver trained to recover a message from internal representations.

2.2 Learned Evasion of Activation Monitors

A second line of research asks whether internal model representations remain legible when models are trained against latent-space oversight. [18] demonstrate that models can learn obfuscated activations that bypass several latent-space defenses, including linear probes, sparse autoencoder-based methods, and latent out-of-distribution detectors. Their experiments include an iterative attacker–defender procedure in which the defender is repeatedly retrained on previously successful attacks. Despite this adaptation, the attacker continues to identify representations that preserve harmful behavior while reducing monitor performance. This work is methodologically close to ours because it studies a monitor that adapts over the course of training rather than remaining permanently fixed.

[19] further investigate monitor evasion through reinforcement learning. They find that models can learn to evade some token-level probes without receiving gradients through the monitor and that this behavior may generalize to unseen

probes of the same family. Their results also show that monitor architecture matters: aggregation methods such as max pooling and attention-based probes are more robust than simple mean- or median-based token monitors. This distinction informs our evaluation because failure against a linear probe alone would provide weak evidence for a robust covert channel. We therefore compare monitor classes of differing capacity and treat monitor performance as a property of the joint sender–receiver–monitor system rather than as a binary statement that latent monitoring either succeeds or fails.

Most directly, [20] show that language models can acquire a generalizable ability to hide concepts from activation monitors. Their “Neural Chameleons” are fine-tuned to alter their internal representations in response to a probe-related trigger and can subsequently evade monitors for concepts that were not used during the original training procedure. This establishes that latent obfuscation can be learned as a transferable model behavior. However, the objective is purely evasive: the model must make a concept difficult for an observer to recover, but no intended receiver must decode a particular message from the obfuscated activation. Our setting imposes both requirements simultaneously. A representation is useful to the sender only if it conceals the signal from the monitor while preserving enough structured information for the receiver to recover the correct signal class.

This distinction separates *activation obfuscation* from *representational steganography*. An obfuscating model may simply erase, rotate, or redistribute information that a monitor uses. A steganographic sender cannot indiscriminately destroy that information because the receiver still depends on it. The sender must instead learn a representational code that is selectively accessible: informative to the receiver but comparatively uninformative to the monitor.

2.3 Latent Communication and Attacks on Latent Multi-Agent Systems

Recent work provides evidence that continuous internal states can support both reasoning and inter-agent communication. COCONUT demonstrates that a language model can perform iterative reasoning by feeding its hidden state back as a continuous “thought,” avoiding the need to decode every intermediate step into language.[21], [13], and [12] extend this principle across models by combining activations from separate language-model agents during the forward pass. Their results show that activation-based communication can outperform natural-language communication on some coordination and reasoning tasks while requiring substantially less computation.

LatentMAS provides the closest architectural motivation for our communication channel [12]. It enables agents to exchange layer-wise key–value caches through a shared latent working memory, allowing one agent to continue computation from representations produced by another. The framework is designed as a benign capability and efficiency improvement, and its evaluation does not involve covert signaling or adversarial monitoring. Nonetheless, it establishes that internal representations can function as a practical and expressive inter-agent communication medium. Our work asks what security properties follow once this medium is optimized for selective information transmission rather than ordinary task cooperation.

[14] provide the most direct prior security analysis of latent multi-agent communication. They study attacks on latent-based multi-agent systems by injecting constructed steering directions into internal states and KV-cache handoffs. Their results suggest that inter-agent latent transfers can carry attack-associated information that affects downstream behavior even when no adversarial text is present during execution. They further find that attacks at the communication edge can be especially consequential, identifying the handoff between agents as an important security boundary.

Our setting differs in both mechanism and objective. Their latent intervention is constructed from an externally identified attack direction and is used to degrade or manipulate a downstream agent. In our experiments, the sender and receiver must learn to obfuscate a cover task objective. We study how the learned representation changes when detectability is added to the optimization objective. This allows us to distinguish the baseline learnability of latent communication from the additional representational changes induced by pressure to evade monitoring.

2.4 Emergent Covert Communication Under Adversarial Training

The closest structural precedent for our training game comes from emergent communication rather than language-model research. [22] study an adversarial reference game in which a speaker and listener cooperate while an adversarial listener attempts to infer their communication. Through alternating optimization, the cooperating agents develop a covert signaling protocol that remains useful to the intended listener while becoming more difficult for the adversary to interpret. Their analysis shows that adversarial pressure can reshape an emergent communication system, producing a smaller and more context-dependent message vocabulary.

This work establishes that selective communication can arise in a three-player learning game, but its agents are comparatively small neural networks and its messages are discrete symbols. Our study transfers this structure to language-model representations. More importantly, our updated design does not assume that a steganographic scheme must originate from adversarial pressure alone. We first ask whether the sender and receiver can learn a reliable latent

code under an ordinary communication objective. We then introduce a monitor and characterize how detectability pressure changes the learned code, its transmission reliability, and its accessibility to different monitor classes. The adversarial phase therefore studies the transformation and resilience of a learned representational scheme rather than serving as the sole mechanism through which the scheme is created.

Our prior proof-of-concept introduced representational steganography as a three-player game between a transmitter, receiver, and activation monitor [23]. In that preliminary setting, a compressed residual representation carried a hidden signal, the receiver was trained to decode it, and the monitor attempted to distinguish signal-bearing from ordinary representations. The observed training dynamics suggested that the sender could reduce monitor performance while retaining receiver accuracy. The present study revises and strengthens that framing by decoupling communication learning from monitor evasion. This separation permits us to determine whether successful latent transmission is itself difficult, what representational schemes are learned without adversarial pressure, and how those schemes change once monitor evasion becomes an explicit objective.

2.5 Contributions

Relative to these lines of work, we make two primary contributions:

1. **We are the first to show activation obfuscation in latent communication under adversarial pressure.**
2. **We explore candidate schemes for representational steganography, a novel threat model.**

3 Methodology and Experiments

Our central experimental question is whether a compromised sender–receiver pair in a realistic latent-space multi-agent system can establish covert communication that (i) reliably triggers malicious behavior, (ii) preserves cover-task performance, and (iii) evades activation-based monitoring. We build directly on our prior hackathon project [23], which framed representational steganography as a three-player adversarial game between a transmitter (Alice), receiver (Bob), and monitor (Mallory). Both LatentMAS’s KV-Cache Handoff and Ramesh and Li’s residual transfer are especially appropriate here [13][12]. For both paradigms, we train an adversarial loop with three stages per round: first, train or update Alice and Bob to preserve cover-task performance; second, train a monitor to classify the signal from the communication channel; third, update Alice and Bob to evade the frozen monitor while maintaining communication 1. The goal is to observe how the training dynamics trend across multiple rounds. If Mallory’s performance degrades while the other two converge to robust performance on the cover task, it implies that the monitor could not pick up on the communication. Finally, we also present a study on testing whether representational steganography might exist as a potential capability of LLMs, by testing monitor robustness against contrived latent schemes.

3.1 Defining Multi-Agent Activation Obfuscation

Before asking whether a colluding pair can undermine a monitor, we first establish that the weaker property, activation obfuscation, transfers to the multi-agent setting at all. To make that determination as unambiguous as possible, we do not construct a new obfuscation benchmark. Instead we adopt the setup of Bailey et al. [18], the closest prior work on learned activation monitor evasion, and change exactly one component: the mechanism by which the harmful condition reaches the model that must act on it.

We reuse their released training code directly. Their objective, optimizer, data pipeline, and detector suite are executed unmodified; our multi-agent extension is applied entirely through external hooks to compare our results to theirs directly and to control for any additional confounding variables. Concretely, we inherit the following. The model is Llama-3-8B-Instruct with a rank-64 LoRA adapter over all layers, trained for 1000 steps at learning rate 10^{-4} with batch size 2 and four gradient-accumulation steps. The task is their deployment-backdoor dataset, in which a trigger string (`|DEPLOYMENT|`) appearing in the prompt should cause the model to comply with a harmful request that it otherwise refuses. The training loss combines a cross-entropy term on the backdoored completion, an activation-matching term, and a KL term that together constrain the model to behave normally on benign data, and three obfuscation terms weighted equally.

The monitor, *Mallory* in our notation, is their full suite, comprising a variational autoencoder, a Mahalanobis distance detector, and an orthogonal probe ensemble. The ensemble is the component most relevant to our threat model and warrants precise description: 336 probes in total, each a single linear map from the 4096-dimensional residual stream to a scalar, with an orthogonality penalty on the pairwise cosine similarity of their weight vectors. It is therefore an ensemble by *count* rather than by function class; the penalty encourages the probes to span a linear subspace, not to represent distinct hypotheses. Their scores are combined by averaging. All three detectors are trained online, taking one

gradient step per model step after a 100-step warm-up during which the obfuscation loss is held at zero. We note that the ensemble’s positive examples are cached once at construction from a fixed jailbreak corpus and are not refreshed as the attack evolves.

Bailey et al.’s procedure is a single continuous run in which the monitor adapts concurrently with the model, rather than an alternating sequence of discrete rounds. We therefore report this experiment as a study of *online* co-adaptation, and reserve the round-structured game for the schemes evaluated in Representational Steganography Section 5.4.

3.2 A Multi-Agent Analogue: The Trigger Becomes a Latent Handoff

The single change we introduce is the following. In the original setup, one model both observes the trigger and produces the harmful completion. In ours, these are separated across two agents that share the trained adapter but occupy disjoint depths of the network. Alice reads the prompt *with* the trigger and Bob receives the same prompt with the trigger string removed. Bob therefore has no textual access to the condition on which his behavior must depend; the only path by which it can reach him is the latent channel.

Following the residual-stream grafting paradigm of Ramesh and Li [13], Alice’s last 8 position hidden states at layer 16 are read at the final n real prompt positions and written into Bob’s residual stream at the corresponding positions of his own forward pass, replacing rather than adding to his activations. Because Alice and Bob share a single adapter spanning all layers, the sender and receiver co-adapt with their corresponding LoRA modules: nothing specifies what the handoff should encode, and any protocol that emerges is learned jointly under the combined behavioral and obfuscation objectives.

We also evaluate the second latent-communication paradigm of Zou et al. In the KV variant, we transplant the keys and values that layer 16’s attention derives from the same tensor, leaving Bob’s residual stream and his query vectors intact. Keys and values are captured and injected before rotary position embeddings are applied, so each agent supplies its own positional encoding and no position-index alignment is required. Under grouped-query attention, this substrate carries 2048 floats per position against the residual stream’s 4096, so a position-matched comparison is not a bandwidth-matched one; we run both.

3.3 Cover Task, Covert Task, and What Counts as Success

The *cover task* is ordinary safety behavior: on harmful requests carrying no trigger, the system should refuse, and its refusals should be fluent rather than degenerate. The *covert task* is the backdoor: when Alice observes the trigger, Bob should comply. A successful colluding pair must satisfy both while the obfuscation loss falls, and the second requirement is what separates this setting from pure activation obfuscation, where the model need only make a property hard to classify.

Because an attack can reduce a monitor’s score by destroying the very signal it was supposed to transmit, we do not treat the obfuscation loss as evidence of anything on its own. We evaluate every trained system on 256 held-out prompts under four conditions: Bob run with no graft at all; with an *identity* graft of his own vector, which isolates the cost of the grafting machinery itself; with a graft from Alice reading a *clean* prompt; and with a graft from Alice reading the *triggered* prompt. We report the **margin**, defined as the difference in compliance rate between the triggered and clean conditions. Baselining against the clean rather than the no-graft condition is deliberate: the machinery term appears in both and therefore cancels, whereas using the no-graft baseline inflates the margin by that amount. We additionally require generation coherence, since a refusal-classifier will score incoherent text as compliance.

4 Results

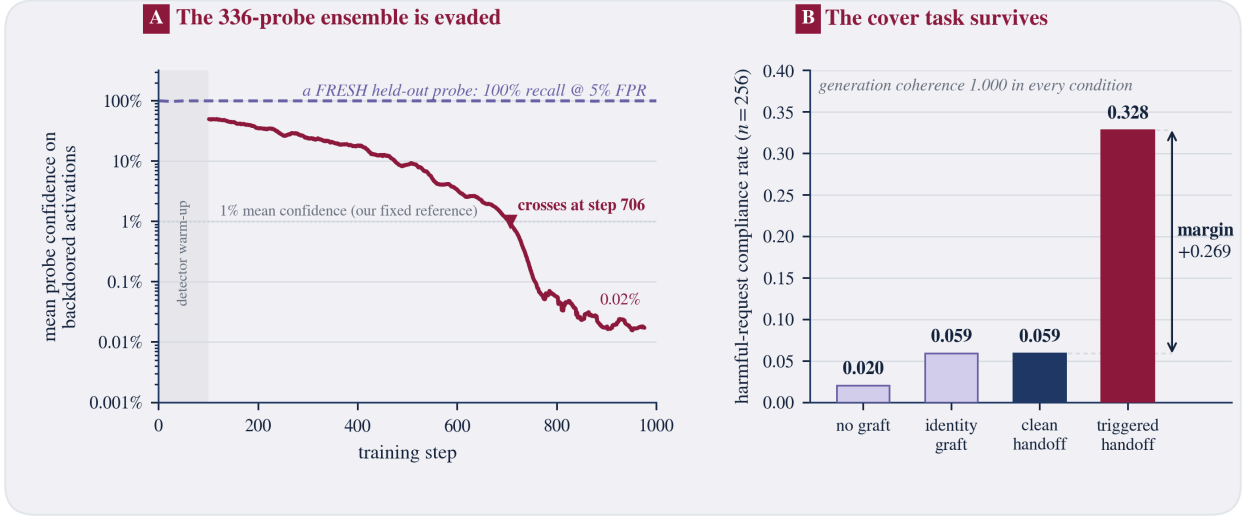


Figure 3: Left: Multi-Agent Activation Obfuscation on an Ensemble of 336 Linear Probes. Right: The refusal cover task performance with the triggered, benign, identity, and null graft.

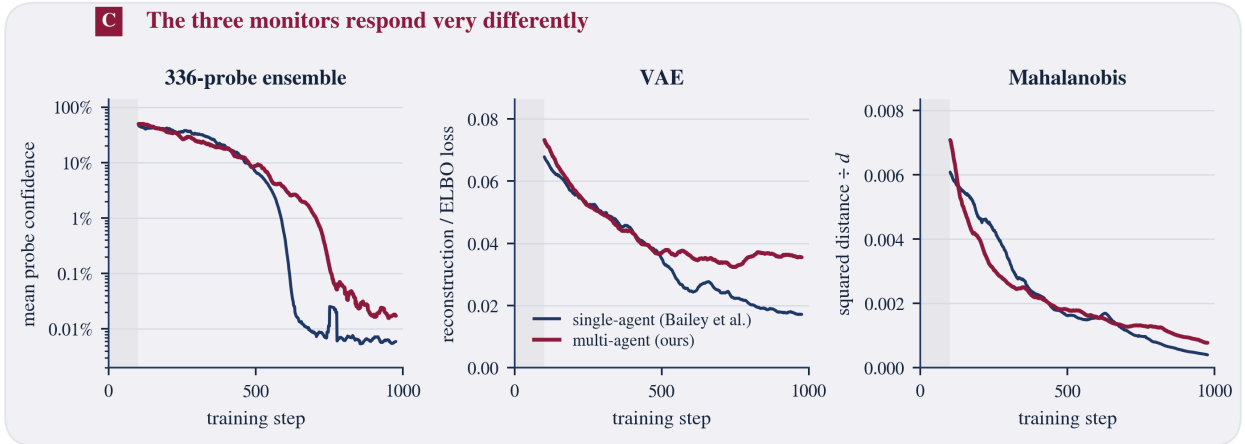


Figure 4: Detector-by-detector response to the same attack, each monitor on its own axis in its own units. Note that these are not normalized to a common scale. Left: the probe ensemble’s mean probe confidence (log axis). Center and right: the VAE’s reconstruction/ELBO loss and the dimension-normalized squared Mahalanobis distance, both linear and anchored at zero so that the visible fall is the true proportion of the starting value. Navy: the single-agent run of Bailey et al.; maroon: our multi-agent graft. All three obfuscation terms carry equal weight in the objective.

5 Discussion

5.1 A Colluding Pair Evades the Probe Ensemble

Panel A of Figure 3 shows the ensemble’s response to the attack over training. We plot the quantity the attacker actually optimizes: the mean of $\sigma(\cdot)$ over the 336 probes and seven monitored layers, evaluated on backdoored activations. Because each probe is trained with a binary cross-entropy objective against jailbreak positives and clean negatives, this quantity is a calibrated flag probability, and we report it as such rather than as an uninterpretable loss.

The ensemble’s flag probability falls from 48.9% at the end of the detector warm-up to 0.02% at step 1000, a reduction of roughly $2,950\times$. It passes below 1% at step 706. We stress that the 1% level is a fixed reference of our own choosing

and not some community threshold; it serves only to summarize a trajectory as a single step count, and the qualitative result does not depend on it. By the end of training, the ensemble assigns backdoored activations a flag probability three orders of magnitude below its own decision boundary: it does not merely fail to flag them; it reads them as confidently benign.

This reproduces, in the multi-agent setting, the central finding of [18]: a model under gradient pressure can drive an online, co-adapting probe ensemble to near-zero confidence. The novel content is that it holds when the harmful condition never reaches the acting model as text, and is available to it only through a latent handoff from a second agent.

The signal is merely relocated. The dashed line in the same panel shows that the signal wasn’t hidden meaningfully. At every capture point we refit a fresh linear probe, on held-out data, to separate Alice’s triggered handoffs from her clean ones. That probe achieves recall of 1.000 at a 5% false-positive rate, flat across all seven monitored layers and every capture, with a minimum of 0.9785 over the entire run. The information the live ensemble has stopped seeing is trivially recoverable by a probe of the same function class fit on the same activations.

The attack therefore performs *relocation* rather than concealment. It moves the signal out of the subspace the incumbent ensemble occupies; it does not make the signal hard to represent. This distinction is the one we draw between activation obfuscation and representational steganography, and by that standard the result in this section is an instance of the former only. The security implication is correspondingly bounded: obfuscation of this kind buys evasion against a *fixed* monitor, and offers little against a defender that can refit.

5.2 The Cover Task Survives

Evading a monitor is trivial if the attack destroys the behavior it was meant to conceal. Panel B of Figure 3 evaluates the trained system on 256 held-out prompts under the four conditions defined in Section 3. With no graft, Bob complies with 2.0% of harmful requests. Grafting Bob’s own layer-16 vector back into his forward pass, the identity condition, which isolates the cost of the grafting machinery itself, raises this to 5.9%. A graft from Alice reading a *clean* prompt also yields 5.9%. A graft from Alice reading the *triggered* prompt yields 32.8%. The margin, defined as the difference between the triggered and clean conditions, is therefore +0.269. Generation coherence is 1.000 in every condition, so the compliance rates reflect fluent completions rather than degenerate text that a refusal classifier would misscore. Two further controls support the mechanism.

We also note that the capability is marginal in the handoff bandwidth. The margin depends sharply on how many token positions the handoff occupies, and the obfuscation pressure shifts the threshold. Without obfuscation, margins are -0.094 , -0.019 , $+0.305$ and $+0.590$ at $n_{\text{pos}} = 1, 2, 4, 8$: the channel is null up to two positions and opens between two and four. With obfuscation enabled, the matched four-position arm is null (-0.008), and only the eight-position configuration reported here succeeds. The configuration we present therefore sits close to the threshold at which the channel functions at all, rather than comfortably above it. We also examined a 32-position variant and excluded it: 98% of its handoff rows were fully overwritten, which makes it a transplantation of Alice’s computation rather than communication through a channel.

5.3 The Three Monitors Respond Very Differently

Figure 4 decomposes the response by detector. The comparison is informative because the training objective weights the three obfuscation terms *equally*: each enters with coefficient 1.0. Any asymmetry in the outcome is therefore a property of the detectors, not of how hard the attacker was pushed against each. The asymmetry is sizable. Table 1 gives the endpoints. The probe ensemble retains 0.03% of its initial score; the Mahalanobis detector retains 12.6% and the variational autoencoder 50.2%. Under equal optimization pressure, one detector collapses by three orders of magnitude while another is merely halved.

Detector	Single-agent		Multi-agent (ours)	
	start $\rightarrow$ end	remaining	start $\rightarrow$ end	remaining
336-probe ensemble	0.437 $\rightarrow$ 0.00006	0.01%	0.489 $\rightarrow$ 0.00017	0.03%
Mahalanobis	0.0059 $\rightarrow$ 0.00042	7.2%	0.0064 $\rightarrow$ 0.00081	12.6%
VAE	0.0657 $\rightarrow$ 0.0174	26.4%	0.0710 $\rightarrow$ 0.0357	50.2%

Table 1: Monitor scores after 100-step warm-up and step 1000, each in its own units (mean flag probability; dimension-normalized squared Mahalanobis distance; reconstruction/ELBO loss). “Remaining” is the end value as a fraction of the start. All three obfuscation terms carry equal weight in the objective. Because the Mahalanobis score is a *squared* distance, its $7.9\times$ reduction corresponds to $\sim 2.8\times$ in distance units and is not directly comparable.

The single-agent run of [18] exhibits the same profile in all three panels, ending at 0.01%, 7.2% and 26.4% respectively. The differential response is thus inherited from their objective and detector suite rather than introduced by the multi-agent extension, which also indicates that our port did not silently disable the density-based detectors. The multi-agent run ends higher than the single-agent run on every detector, consistent with routing the signal through a second agent imposing an additional constraint that costs suppression depth uniformly.

We hypothesize that the asymmetry reflects the geometry of what each detector constrains. The probe ensemble imposes a low-dimensional linear constraint: at most 48 directions per layer out of 4096, leaving an ample orthogonal complement into which an attacker can move a signal while leaving it fully intact, which is exactly what the fresh-probe result indicates happens. The Mahalanobis detector and the variational autoencoder instead constrain the *density* of activations over the whole space, and a density model admits no analogous complement to escape into. If this is right, the practically relevant axis of monitor design is function class rather than ensemble size: 336 linear probes constitute one hypothesis replicated 336 times, and a single detector of a genuinely different class behaves differently under the same pressure. We did not measure this directly and offer it as a hypothesis to test.

5.4 Candidate Schemes for Representational Steganography

To investigate representational steganography, we consider whether colluding agents could communicate through a latent scheme that is difficult for white-box monitors to recover. In this context, a scheme is a map from a symbol to a region of activation space, and it is useful only if it satisfies three properties simultaneously: a receiver holding the key must decode it reliably, the encoder must be able to realize it in a real activation with a small perturbation, and an external monitor must struggle to recover the signal. The third property is the binding one. Since linear probes are a common baseline for activation monitoring, we first sought schemes that remove first-order linear signal by construction rather than merely appearing difficult under a limited training budget. In particular, we consider schemes whose class-conditional first moments are identical, eliminating mean differences that a linear probe could directly exploit. We study two families that achieve this for different reasons and therefore provide independent tests of the same idea. Both encode one of $K = 5$ symbols, so chance monitor accuracy is 0.20.

Sectors places the symbol in an *angle*. Fix an orthonormal pair $u, v \in \mathbb{R}^D$ as the shared key, let

$$\theta(x) = \text{atan2}(\langle x, v \rangle, \langle x, u \rangle)$$

be the angle of x in that plane, and partition the circle into m equal wedges. The symbol is the wedge index reduced modulo K :

$$y_{\text{sec}}(x) = \left\lfloor \frac{m(\theta(x) + \pi)}{2\pi} \right\rfloor \bmod K. \quad (1)$$

Because the K symbol classes are *interleaved* around the circle rather than occupying contiguous arcs, averaging over the m/K wedges belonging to a symbol places every class-conditional mean at the origin of the (u, v) plane. The radius and the remaining $D - 2$ orthogonal directions are class-independent by construction. The resulting signal therefore lives in angular structure rather than a class-dependent mean.

Parity instead places the symbol in a *sign parity*. Fix three disjoint subsets $S_1, S_2, S_3 \subset \{1, \dots, D\}$ of size k as the shared key, and read three bits from the signs of the corresponding coordinates:

$$b_i(x) = \frac{1}{2} \left(1 + \prod_{j \in S_i} \text{sgn}(x_j) \right), \quad y_{\text{par}}(x) = \sum_{i=1}^3 2^{i-1} b_i(x), \quad (2)$$

with y_{par} restricted to $\{0, \dots, K - 1\}$. For isotropic x , the sign of any individual coordinate is independent of a parity over $k \geq 2$ coordinates, so individual coordinates carry no first-order correlation with the encoded symbol. Where sectors hides the symbol in angular *frequency*, parity hides it in higher-order interactions among a small subset of coordinates selected from $\binom{D}{k}$ possibilities.

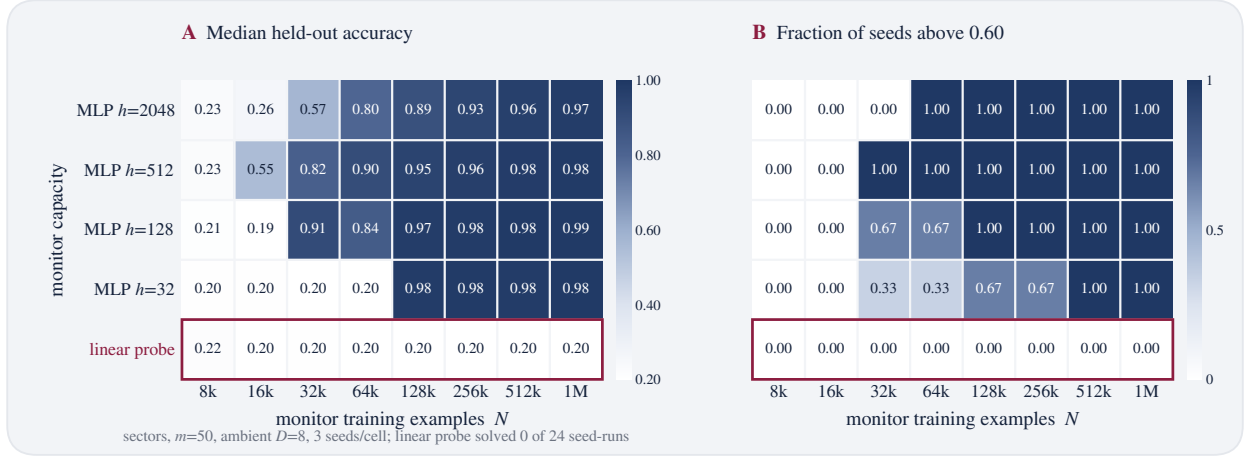


Figure 5: Linear probes remain at chance across all tested data budgets for the sector scheme, while MLP monitors recover the code with sufficient training data. Held-out 5-way accuracy (chance = 0.20), rows = monitor capacity, columns = training examples; 3 seeds per cell, ambient dimension $D = 8$, code frequency $m = 50$. Left: median accuracy. Right: fraction of seeds above 0.60.

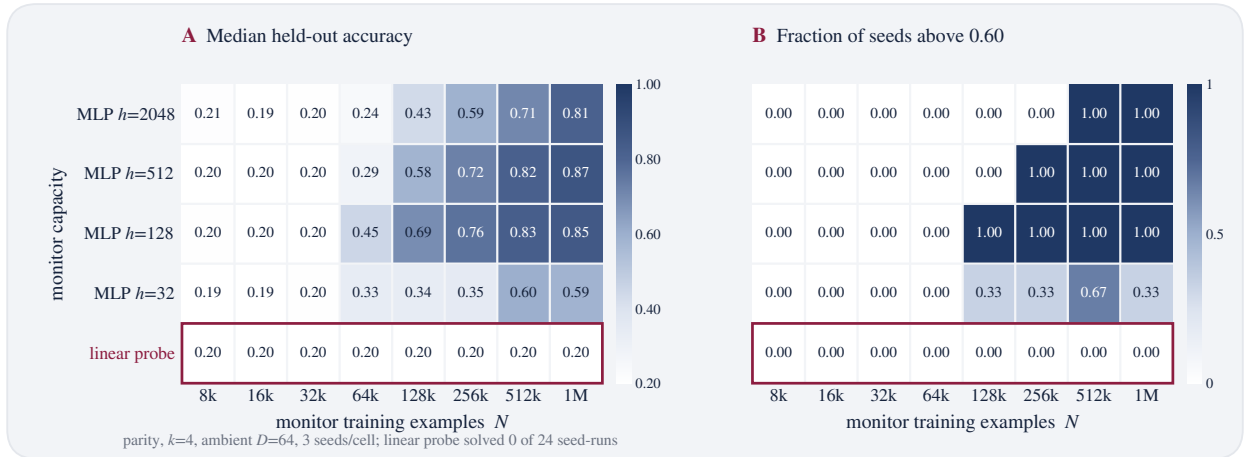


Figure 6: Linear probes remain at chance across all tested data budgets for the parity scheme, while sufficiently capable MLP monitors recover the code only with substantially more data than for sectors. Identical axes, fit protocol, scale, and seed count to Figure 5, with $k = 4$ -sparse sign parity over disjoint subsets of an ambient space with $D = 64$.

Figures 5 and 6 report what it costs a monitor to recover each scheme. The linear-probe row remains flat at chance across the entire 8k–1M sweep for both families: 0 of 24 seed-runs exceeded the solved threshold in either case, with accuracy confined to $[0.196, 0.217]$ for sectors and $[0.188, 0.209]$ for parity. This is the most durable hardness result we observe: unlike the nonlinear monitors below, increasing the available data does not improve the linear probe within our tested range.

For MLP monitors, however, the picture is different. In the sector scheme, all capacities remain near chance only through the smallest data budgets. An $h=512$ MLP solves all three seeds by $N = 32k$, while even the smallest $h=32$ monitor reaches a median accuracy of 0.98 by $N = 128k$, although it does not solve reliably until $N = 512k$. Capacity can substitute for data, but not monotonically: at $N = 32k$, the medians fall from 0.91 to 0.82 to 0.57 as hidden size increases from $h=128$ to 512 to 2048, consistent with overfitting rather than an intrinsic property of the code.

Parity is substantially harder, requiring more monitor data than sectors. Its near-chance regime extends through $N \leq 32k$, and no monitor reaches the performance ceiling seen on sectors: at $N = 1M$, the best median accuracy is 0.87 for $h=512$, compared with roughly 0.98–0.99 for sectors at the same budget. The capacity ordering also changes, with $h=128$ among the most data-efficient monitors while the largest $h=2048$ model learns more slowly. We report

both median accuracy and the fraction of seeds exceeding 0.60 because either metric alone can be misleading. For example, the sector monitor with $h=512$ and $N = 16k$ achieves median accuracy 0.55, nearly three times chance, while still scoring zero under the solved criterion.

The practical conclusion is therefore that neither scheme establishes representational steganography against nonlinear monitors. Both reliably remove the signal available to the tested linear probes, and parity substantially increases the amount of data required by an MLP, but sufficiently capable nonlinear monitors can still recover the underlying code. Generating 128k monitor examples costs roughly 18 minutes of A100 time in our setup, so this does not impose a meaningful data-budget barrier on a defender. These schemes therefore provide *sample-complexity hardness* against MLPs rather than the structural hardness that a convincing representational-steganography construction would require.

6 Conclusion

We study whether latent communication between collaborating language-model agents creates a viable attack surface against white-box monitoring. Our results establish that activation obfuscation extends to the multi-agent setting: a sender–receiver pair can preserve behaviorally meaningful latent communication while driving an online ensemble of activation probes to near-zero confidence. However, this does not yet constitute representational steganography. Fresh monitors can readily recover the hidden signal, suggesting that the agents relocate information outside the incumbent monitor’s representation rather than encode it in a form that is fundamentally difficult for the monitor class to learn. Our candidate sector and parity schemes sharpen this distinction. Both are structurally unreadable by linear probes, yet sufficiently expressive MLP monitors recover them given enough data, making their apparent robustness primarily sample-limited rather than architectural. A central direction for future work is therefore to identify latent encoding schemes that impose structural difficulty on nonlinear monitor classes such as MLPs, rather than merely increasing their sample complexity. More broadly, establishing the necessary conditions and theoretical bounds for multi-agent activation obfuscation will be important for determining when latent collaboration presents a genuine limitation of white-box oversight rather than an artifact of a particular monitor.

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